

Python code:

```
import cv2
import mediapipe as mp
import serial
import time

arduino = serial.Serial('COM17', 9600)
time.sleep(2)

mp_hands = mp.solutions.hands
hands = mp_hands.Hands(max_num_hands=1,
min_detection_confidence=0.7)
mp_draw = mp.solutions.drawing_utils

cap = cv2.VideoCapture(0)

last_sent = -1

while True:
    success, img = cap.read()
    img = cv2.flip(img, 1)
    imgRGB = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)
    results = hands.process(imgRGB)

    finger_count = 0

    if results.multi_hand_landmarks:
        for handLms in results.multi_hand_landmarks:
            mp_draw.draw_landmarks(img, handLms,
mp_hands.HAND_CONNECTIONS)

            tips = [8, 12, 16, 20]

            for tip in tips:
                if handLms.landmark[tip].y < handLms.landmark[tip - 2].y:
                    finger_count += 1
```

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        if handLms.landmark[4].x > handLms.landmark[3].x:
            finger_count += 1

# send only when changed
if finger_count != last_sent:
    arduino.write(str(finger_count).encode())
    print("Sent:", finger_count)
    last_sent = finger_count
    time.sleep(0.2)

cv2.putText(img, f'Fingers: {finger_count}', (20, 50),
            cv2.FONT_HERSHEY_SIMPLEX, 1, (255, 255, 0), 2)

cv2.imshow("Gesture System", img)

if cv2.waitKey(1) & 0xFF == 27:
    break

cap.release()
cv2.destroyAllWindows()
arduino.close()
```